SC-SCOPE endpoint floor sharpening – the reconciliation (proved) and

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-08 · this version issued 2026-06-08 · v1.5

1. Purpose and scope

The SC-SCOPE all-orders endpoint ($I = 2 \times 10^{-3}$) did not close by third-order means: the joint incompatible-pairing recovered only $\times 0.905$ (scscope-joint-pairing v1.0), which named the SOLE remaining route as "a larger second-order STEP-5B endpoint floor: if the de-thinned floor were $\gtrsim \times 3.9$ the paired joint would close." This note completes that route by observing that the floor $\rho = 2.58$ rests on a LOOSE additive-energy bound, and the additive-energy result of the DR-2 programme (R-025/R-026) supplies the tighter constant.

2. The floor composition

The de-thinned STEP-5B closure floor (robustness_mu2_step5b_remargin) is

$$\rho = \frac{K_{\text{budget}}}{K(n_{\text{pack}})}, \qquad K_{\text{budget}} = \frac{4(1-a_0)\,m}{(\lambda' I)^2 J_{\text{eff}}}, \qquad K(n_{\text{pack}}) = 8 + 4\sqrt{14}\,\sqrt{n_{\text{pack}}},$$

where $K(n_{\text{pack}})$ is the kappa-balanced additive-energy bound and $n_{\text{pack}} = 16/\theta^2_{\text{min}}$. At the endpoint (anchor $\mu^2 = 0.005$): $n_{\text{pack}} = 40.7$, $K(n_{\text{pack}}) = 103.5$, $K_{\text{budget}} = 266.7$, so $\rho = 2.58$ (reproducing the AddE $\times 2.6$).

3. The bound is loose at the small endpoint

$K(n) = 8 + c_R\sqrt{n}$ with $c_R = 4\sqrt{14} \approx 15$ is the UNCONDITIONAL triple-count bound ($E_+ \le n^{5/2} \Rightarrow K = E_+/n^2 \le \sqrt{n}$ up to the prefactor). At the endpoint $n_{\text{pack}} = 40.7$ this gives 103.5, which EXCEEDS the trivial Lemma-A additive-energy bound

$$\frac{E_+(Q)}{N^2} \le 1 + T'(Q), \qquad T'(Q) \le |Q| \le n_{\text{pack}} = 40.7 \quad \Rightarrow \quad 1 + T' \le 41.7 < 103.5.$$

The large prefactor $c_R \approx 15$ makes the $\sqrt{n}$ bound overshoot the trivial $O(n)$ bound at this moderate n. The additive-energy constant K and $1 + T'$ are the SAME λ'-free quantity $(\langle F^4 \rangle - 4I^2)/I^2$ (via $\sum w^2 = \lambda'^2(\langle F^4 \rangle - 4I^2)$ and $\sum w^2 \le K(\lambda' I)^2$), so $1 + T'$ is a legitimate, tighter substitute.

4. The sharpened floor: a partial advance, NOT a closure

Correction (v1.3). v1.1/v1.2 here computed the closure as paired $= \rho_{\text{lat}}/(1+\max[R_s+R_q]) = \rho_{\text{lat}}/2.872$ and concluded the endpoint LIFTS. That formula's linear-in-ρ growth is only a LOCAL approximation at $\rho = 2.6$. The physically-correct accounting (scscope_joint_endpoint) treats the third-cumulant SUNSET as an ABSOLUTE cost $C_{\text{sunset}} = \text{composed}/1.13$ that does NOT vanish as the second-order floor thickens; the joint ratio therefore SATURATES:

$$\text{joint}(\rho) = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}} \xrightarrow{\rho\to\infty} \times 1.13,$$

NOT linearly. At the sharpened floor ($K \le T' \le n_{\text{pack}} = 40.7 \Rightarrow \rho_{\text{lat}} = 6.55$) the additive joint is only $\times 0.945 < 1$; the verified $K \le 0.52\,T'$ gives $\rho_{\text{lat}} = 12.6$ and $\times 1.026$ – MARGINAL. The threshold for joint > 1 is $\rho \ge 9.85$ (i.e. $K \le 27$), not the 3.9 the lift used (scscope_joint_correction.py, 5/5). So the proved floor sharpening is a real PARTIAL advance – it moves the additive endpoint joint from $\times 0.757$ to $\times 0.95$–1.03 – but does NOT cleanly close the endpoint.

5. Reconciliation (PROVED) and the lift

The constant map is now pinned EXACTLY. In the STEP-5B convention $w_t = \lambda' \sum_{u+v=t} A_u A_v$ and, by the Parseval reformulation, $\sum_{t\neq 0} w_t^2 = \lambda'^2(\langle F^4\rangle - 4I^2)$, so the floor's additive-energy constant is exactly $K = \sum_{t\neq 0} w_t^2/(\lambda' I)^2$. Writing $v_t = \sum_{u+v=t} A_u A_v$ (so $w_t = \lambda' v_t$), the weighted Lemma A (R-027) applied to the λ'-free v_t gives, for its $t \neq 0$ part, $\sum_{t\neq 0} |v_t|^2 \le T'(M)\, I^2$; hence $\sum_{t\neq 0} |w_t|^2 = \lambda'^2 \sum_{t\neq 0} |v_t|^2 \le T'(M)(\lambda' I)^2$ and

$$\boxed{K = \frac{\sum_{t\neq 0} w_t^2}{(\lambda' I)^2} \;\le\; T'(M)}$$

where $T'(M)$ is the sum-circle richness of the mode set – NOT the looser kappa-balanced $8 + c_R\sqrt{n}$. Verified (scscope_constant_map.py, 3/3) on real $\pm k$-symmetric sphere-lattice patterns: $K_{\text{floor}}/T' \le 0.52$ and $w_0 = I$ exactly (so the $-4I^2$ subtraction is even more conservative, lowering K further). With $K \le T' \le n_{\text{pack}} = 40.7$ (separated) or $T' \sim$ tens (lattice), $\rho_{\text{lat}} = K_{\text{budget}}/K \ge 266.7/40.7 = 6.55$. This is a real, PROVED sharpening of the second-order floor; but per §4 it does NOT close the endpoint under the correct additive bookkeeping (it is marginal, $\times 0.95$–1.03). The reconciliation $K \le T'(M)$ STANDS; the LIFT does not.

Grade (v1.3 corrected). The reconciliation $K \le T'(M)$ is PROVED and the second-order floor is genuinely sharpened ($\rho : 2.58 \to 6.55$). But §4 shows this does NOT close the all-orders endpoint: under the correct additive bookkeeping the sunset saturates the joint at $\times 1.13$, and at the sharpened floor the joint is only $\times 0.945$–1.026 (marginal). The LIFT claimed in v1.1/v1.2 is therefore RETRACTED. SC-SCOPE is RESTORED as a B1 named hypothesis ($\{$H-LAYER$\} \to \{$H-LAYER, SC-SCOPE$\}$), B1 tier UNCHANGED T6. This note's durable content is the PROVED floor sharpening $K \le T'(M)$ and the resulting PARTIAL advance ($\times 0.757 \to \times 0.95$–$1.03$); a clean lift needs either $\rho \ge 9.85$ ($K \le 27$, tighter than $n_{\text{pack}} = 40.7$) or a rigorous third-cumulant bound beating the saturating sunset.

5b. The realized quartic (strong evidence, not certified)

Under the canonical additive bookkeeping the endpoint closes at $\rho_{\text{lat}} = 6.55$ iff the quartic-difference per-transfer ratio satisfies $R_{\max} < 0.634$ (the sunset is rigorous and caps the joint at $\times 1.13$). The prior $R_{\max} = 1.019$ inherited the Young-ceiling estimate $R_{\sup} = 1.59$ (flagged "not recomputed"). Evaluating the realized ratio DIRECTLY,

$$R(t) = 12\Big(\tfrac{5v}{2}\Big)^2 \lambda'^{-2}\,\widehat{G^4}(t)\,\frac{4(1-a_0)}{J(t)}, \qquad \widehat{G^4} = (J * J)(t),$$

on the endpoint chords (scscope_quartic_realized.py, 4/4) gives

$$\boxed{R_{\max}^{\text{realized}} \approx 0.385 \; < \; 0.634}$$

– the Young ceiling was loose by $\sim 2.6\times$. This is STRONG EVIDENCE that the SC-SCOPE all-orders endpoint CLOSES. It is NOT certified: the absolute $\widehat{G^4}$ normalisation carries a factor-$2/(2\pi)^3$ convention (the same class as the $M' = -J(0)$ vs $-J(0)/2$ slip), which is load-bearing here – the closure survives a +50% slack ($0.385 \times 1.5 = 0.577 < 0.634$) but NOT a full factor 2 ($0.769 > 0.634$). The convolution SHAPE is rigorous (grid-converged); only the absolute normalisation is open. NO lift is enacted; SC-SCOPE stays a B1 named hypothesis pending the convention being pinned.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"The reconciliation $K \le T'(M)$ might be wrong."* DISMISSED: it is proved (Parseval + Lemma A $t \neq 0$ part) and verified ($K_{\text{floor}}/T' \le 0.52$, $w_0 = I$; scscope_constant_map.py 3/3). This is the note's durable content and is NOT retracted.

(β) *"Then why is the endpoint not closed?"* Because the third-order joint is SUNSET-LIMITED. Under the canonical additive bookkeeping the sunset cost $C_{\text{sunset}} = \text{composed}/1.13$ does not shrink with the floor, so joint$(\rho) \to \times 1.13$ as $\rho \to \infty$. At $\rho_{\text{lat}} = 6.55$ the joint is only $\times 0.945$ (with the current quartic $R_{\max} = 1.019$); even a perfect quartic gives at most $\times 1.108$. So the floor sharpening alone cannot close it – it is a partial advance.

(γ) *"The earlier paired$= \rho/2.872$ closure was wrong."* CONCEDED – this is the v1.3 retraction. That formula's linear-in-ρ growth is a local approximation at $\rho = 2.6$; the physically-correct accounting (the third-order sunset is an absolute cost) saturates. The retraction is logged at negative-results AUDIT-2026-06-08-scscope-lift-overclaim.

(δ) *"Is $\rho = 2.58$ reproduced?"* Verified: $K_{\text{budget}}/K(n_{\text{pack}}) = 266.7/103.5 = 2.58$ matches the AddE/remargin $\times 2.6$.

Quantitative sanity check (mandatory). Reconciliation: $K \le T'(M)$, $K_{\text{floor}}/T' \le 0.52$, $w_0 = I$ (scscope_constant_map.py 3/3). Floor: $\rho : 2.58 \to 6.55$ (proved sharpening). Joint (CORRECTED, additive bookkeeping): $\times 0.945$ at $\rho_{\text{lat}} = 6.55$; threshold $\rho \ge 9.85$; saturates at $\times 1.13$ (sunset-limited); the quartic must reach $R_{\max} < 0.634$ to close at $\rho_{\text{lat}} = 6.55$ (scscope_joint_correction.py 5/5). Direction: tighter $K \Rightarrow$ larger $\rho \Rightarrow$ smaller C_2 only; the sunset is the binding term. No closure claimed.

7. Result footer

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Result ID:                R-029 / SC-SCOPE endpoint floor sharpening (PARTIAL;
                          lift retracted)
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Precise statement:	PROVED: the floor's additive-energy constant K=sum_{t!=0}w_t^2/(lambda'I)^2 <= T'(M) (reconciliation, scscope_constant_map.py 3/3), sharpening the endpoint floor rho: 2.58 -> 6.55 (the kappa-balanced K=103.5 was loose at n_pack=40.7). RETRACTED: the resulting all-orders LIFT -- it used paired=rho/2.872 (local approx); the correct additive bookkeeping (sunset absolute) gives only x0.945 (conservative) .. x1.026 (verified 0.52T'), MARGINAL; threshold rho>=9.85, joint saturates at x1.13 (sunset-limited). PARTIAL advance x0.757 -> x0.95-1.03.
Scope:	SC-SCOPE all-orders endpoint (I=2e-3); the named 'sharper floor rho>~3.9' route, now completed.
Dependencies:	robustness_mu2_step5b_remargin (floor); R-025 Lemma A (additive-energy bound 1+T'); R-026 (lattice T'~R^eps); scscope-joint-pairing v1.0 (third-order inflation 2.872).
Evidence grade:	Reconciliation K<=T'(M) PROVED + verified (3/3); floor sharpened rho 2.58->6.55. The all-orders LIFT is RETRACTED (the additive joint is MARGINAL x0.945-1.026, sunset-limited; scscope_joint_correction.py 5/5). Net: a PROVED PARTIAL advance, NOT a closure.
Reproduction command:	python codes/vacuum/scscope_floor_sharpening.py
Expected output:	scscope_constant_map.py 3/3 (K<=T'(M)); scscope_joint_correction.py 5/5 (additive joint x0.945 at rho=6.55; threshold rho>=9.85; saturates x1.13).
Falsification gate:	The reconciliation K<=T'(M) failing (not observed). The lift is already retracted; closing the endpoint needs rho>=9.85 (K<=27) OR a certified quartic R_max<0.634.
Tier before / after:	LIFT RETRACTED. SC-SCOPE RESTORED as a B1 named hypothesis: B1 {H-LAYER}->{H-LAYER,SC-SCOPE}, tier UNCHANGED T6. SC-SCOPE endpoint OPEN/MARGINAL. B5 T5. AUDIT-2026-06-08-scscope-lift-overclaim.
No-overclaim statement:	Only the reconciliation K<=T'(M) and the floor sharpening (rho 2.58->6.55) are claimed (proved). The all-orders endpoint is NOT closed: the additive joint is marginal (x0.945-1.026) and sunset-limited (saturates x1.13). SC-SCOPE remains a B1 named hypothesis.
Next required action:	Pin the absolute Ghat4 normalisation (factor-2/(2pi)^3 convention): the REALIZED quartic R_max~0.385 < 0.634 (scscope_quartic_realized.py 4/4) is strong evidence of closure, load-bearing only on that convention. Pinning it (survives +50%, not x2) would certify the endpoint lift.